

PhD Roadmap Proposal: Secure, Reliable and Efficient Transformer-Based Systems

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1. Context and importance

Deployment setting. I propose to study Transformer-based inference on resource-constrained edge systems, using a Vision Transformer (ViT) for safety-sensitive image analysis as the main experimental case. A secondary language-Transformer model can be used later to test whether the findings transfer beyond vision. Such systems may operate with limited memory and energy, so quantization and other efficiency methods are attractive. At the same time, model parameters are stored in DRAM/SRAM and can be affected by random hardware faults such as soft errors or by intentional bit-flip attacks. The attached work shows that these two causes can lead to very different failure behaviour [3].

The problem matters to AI developers, accelerator and memory designers, and operators of safety-sensitive edge or server systems. Efficiency cannot be considered separately from reliability and security: changing numerical precision, parameter layout, or protection mechanisms also changes which bits and model components are important. The goal is therefore not simply to make a Transformer smaller or faster, but to understand and control the trade-off between *efficiency, reliability and security*.

2. Problem formulation

Research problem: How can quantization and memory-level protection of Transformer-based models be jointly designed so that inference memory use and latency are reduced without increasing vulnerability to random hardware faults and targeted bit-flip attacks?

I will focus on three measurable outcomes: (1) inference quality under clean execution, (2) accuracy degradation under controlled random memory faults, and (3) attack success or accuracy degradation under targeted bit-flip attacks. Efficiency will be measured using model size, memory traffic, latency and, where the platform permits, energy. The research will seek Pareto-efficient configurations rather than assuming that one metric should always be maximized. The central question is whether a protection scheme that is effective against random faults remains effective after the model has been compressed and an attacker selects the most vulnerable bits.

3. Summary of the attached papers

TrojLLM. Xue et al. [4] show that security attacks can remain effective even when an LLM is exposed only through a black-box API. Their method searches for universal triggers and poisoned discrete prompts while preserving clean-task accuracy. This establishes that Transformer security is not only a model-parameter problem; the surrounding inference interface can also be an attack surface. However, it does not study hardware faults or the effect of efficiency-oriented model compression on hardware security.

Does Fault Tolerance Safeguard DNNs Against Bitflip Attacks? This work [3] combines selective hardening, activation restriction and quantization, then evaluates both random fault injection and malicious bit-flip attacks. It shows that the same hardening can improve random-fault resilience while providing limited or inconsistent protection against an intelligent bit-flip attack. This is an important starting point for security-aware fault tolerance, but the experiments use CNNs rather than Transformer architectures.

Effective and Memory-Efficient Alternatives to ECC for Reliable Large-Scale DNNs. This work [2] studies memory protection for large CNNs and ViTs and proposes MSET and CEP as low-overhead alternatives to conventional ECC. A key result is that the vulnerability of floating-point parameters is not uniform: protecting selected important bits can provide strong reliability with much lower overhead. The remaining question is how such protection interacts with efficiency transformations and malicious, targeted bit manipulation.

Your Scale Factors are My Weapon: Targeted Bit-Flip Attacks on Vision Transformers via Scale Factor Manipulation. Wang et al. [1] show that quantized ViTs expose an additional attack surface: scale factors. A change to one scale factor can affect a group of weights, and Flip-S uses this property to obtain targeted attacks with substantially fewer bit flips than previous methods. This directly connects efficiency through quantization

with hardware security. However, the work is primarily an attack study; it does not provide a general joint optimization of efficiency, random-fault reliability and attack resistance.

Collective gap. The four papers [1–4] establish important pieces of the problem: LLM/Transformer security, hardware fault tolerance, low-overhead memory protection, and attacks against quantized ViTs. They do not provide a common framework that measures how an efficiency optimization changes both accidental-fault vulnerability and adversarial vulnerability, and then chooses protection according to both risks. My PhD would address this intersection.

4. Testable hypotheses

1. **H1 – Quantization changes vulnerability concentration.** Lower-bit quantization will reduce memory footprint and inference cost, but it will concentrate the effect of faults into a smaller set of critical bits, including scale-factor bits. Therefore, vulnerability will become more uneven as precision is reduced.
2. **H2 – Random-fault hardening does not imply attack resistance.** Selective protection chosen from random-fault vulnerability will improve resilience to stochastic bit errors, but a targeted attacker that searches for critical unprotected bits will still achieve significant degradation. Security-aware selection will reduce this gap.
3. **H3 – Efficiency and protection can be jointly optimized.** Mixed-precision quantization combined with selective protection of high-impact bits can reach a better efficiency–reliability–security trade-off than uniform precision or uniform protection at the same memory budget.
4. **H4 – Transformer structure provides useful vulnerability signals.** Attention, feed-forward layers, normalization parameters and quantization scales will not have equal vulnerability. Architecture-aware sensitivity measures will predict high-impact locations better than uniform protection and can reduce the cost of fault analysis.

5. Proposed strategy for four years

Threat and fault models. I will distinguish accidental soft errors from malicious bit-flip attacks. The reliability model will inject random single- and multi-bit errors into model memory at controlled bit-error rates. The security model will assume an attacker who knows the model architecture and quantization scheme and can target selected stored bits; where appropriate, a stronger white-box model will be used for controlled experiments. The main security goal will be targeted misclassification or degradation, while avoiding claims about attacks being practical outside the stated assumptions.

Models and platforms. The primary models will be small and medium ViTs that can be run repeatedly on accessible GPU/CPU systems, with FP32/FP16 and INT8/INT4 representations. A small language Transformer will be added as a transfer case. Experiments will initially use software fault injection and memory-level simulation; when suitable hardware is available, the strongest findings will be validated on an edge/accelerator platform. This keeps the work reproducible without making access to a particular accelerator a requirement.

Evaluation. For each model and precision, I will measure clean accuracy, latency, memory footprint, memory traffic and energy when available. Reliability will be measured by accuracy degradation, silent-data-corruption rate and fault sensitivity. Security will be measured by attack success rate, number and location of required bit flips, clean-accuracy retention and robustness across triggers. I will use repeated fault-injection trials, ablation studies and multiple random seeds. Baselines will include no protection, uniform protection, conventional ECC where simulatable, and the attached papers’ relevant methods.

Mitigation methodology. First, I will build Transformer-specific vulnerability maps for weights, scales and other stored parameters. Next, I will formulate joint selection of precision and protection under a fixed memory/latency budget. Candidate methods include mixed-precision quantization, selective bit protection, lightweight redundancy and detection, and attack-aware retraining or parameter reconstruction. Each method will be evaluated against both random and targeted faults. The final objective is a practical design rule: given a model and hardware budget, identify what should be compressed and what should be protected.

Year 1 – Measurement and baseline. Reproduce the main reliability/security trends on ViTs; implement fault injection and controlled bit-flip evaluation; benchmark FP16, INT8 and INT4; build a common dataset of vulnerable bits and layers.

Year 2 – Vulnerability modelling. Study how quantization, scale factors, layer type and model size change vulnerability. Develop Transformer-specific sensitivity measures and validate them against exhaustive or sampled fault injection. Produce a first joint efficiency–reliability–security optimization method.

Year 3 – Security-aware protection. Develop and evaluate selective protection and detection methods against adaptive targeted attacks. Study mixed precision and protection budgets, and test transfer across Transformer architectures and at least one language model.

Year 4 – System validation and thesis. Integrate the best methods into a reproducible prototype, validate on a realistic memory/accelerator setup where possible, perform final cross-model experiments and ablations, release evaluation tools where permitted, and write the thesis and publications.

Potential challenges and responses. Hardware access may be limited, so software injection and memory simulation will provide the main experimental path, followed by hardware validation where possible. Attack results may depend strongly on the threat model; therefore all assumptions and attacker capabilities will be reported explicitly. Finally, security, reliability and efficiency may conflict rather than have a single optimum. I will therefore use Pareto analysis and budget-constrained comparisons instead of optimizing only one metric.

References

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